



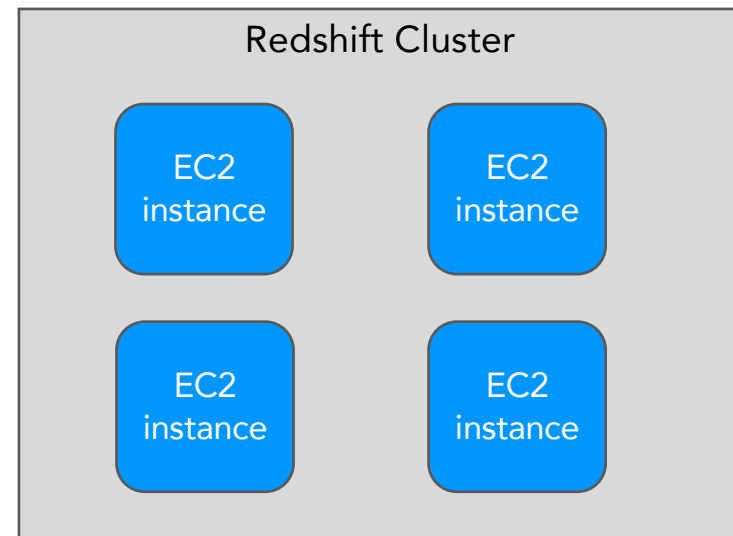
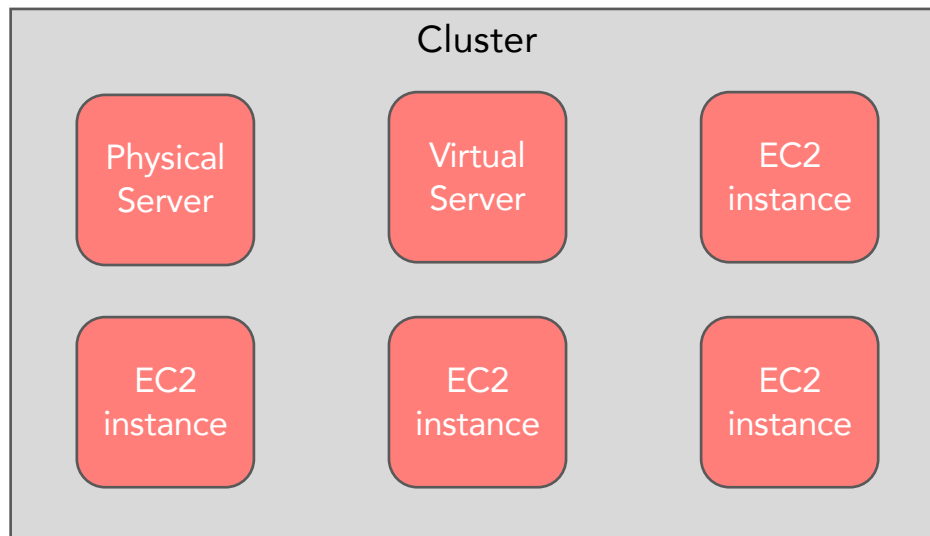
Section 2 – Redshift Infrastructure

What is Redshift

- Data Warehouse on AWS cloud.
 - Data Warehouse
 - Data Mart
 - Data Mesh
- Launched in Feb 2013.
- 11+ years of existence.
- Before Aurora, Redshift was the fastest growing service/product in AWS.
- Amazon Redshift is a **fully managed, petabyte-scale** data warehouse service in the cloud.
- Types of infrastructure offerings
 - Provisioned cluster
 - Serverless

Cluster

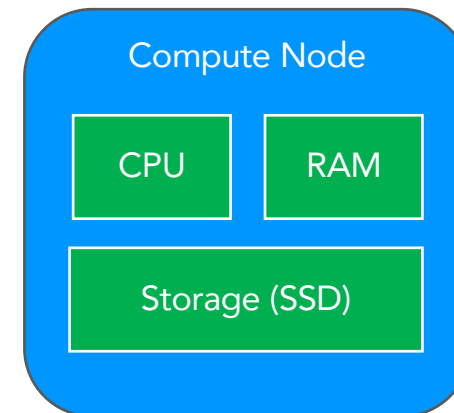
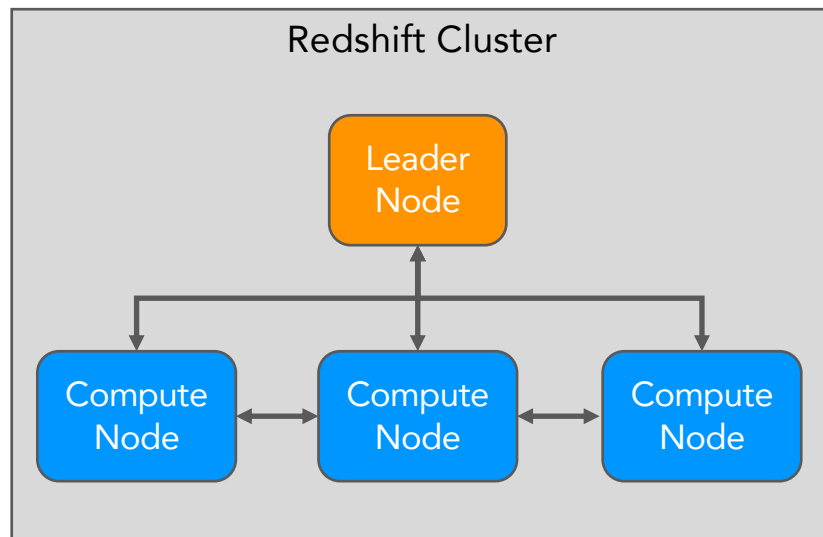
- Group of identical servers (physical/virtual), nodes that work together as a group.
- AWS Console, AWS CLI, AWS SDKs, Redshift API (HTTP)



Redshift AWS Console

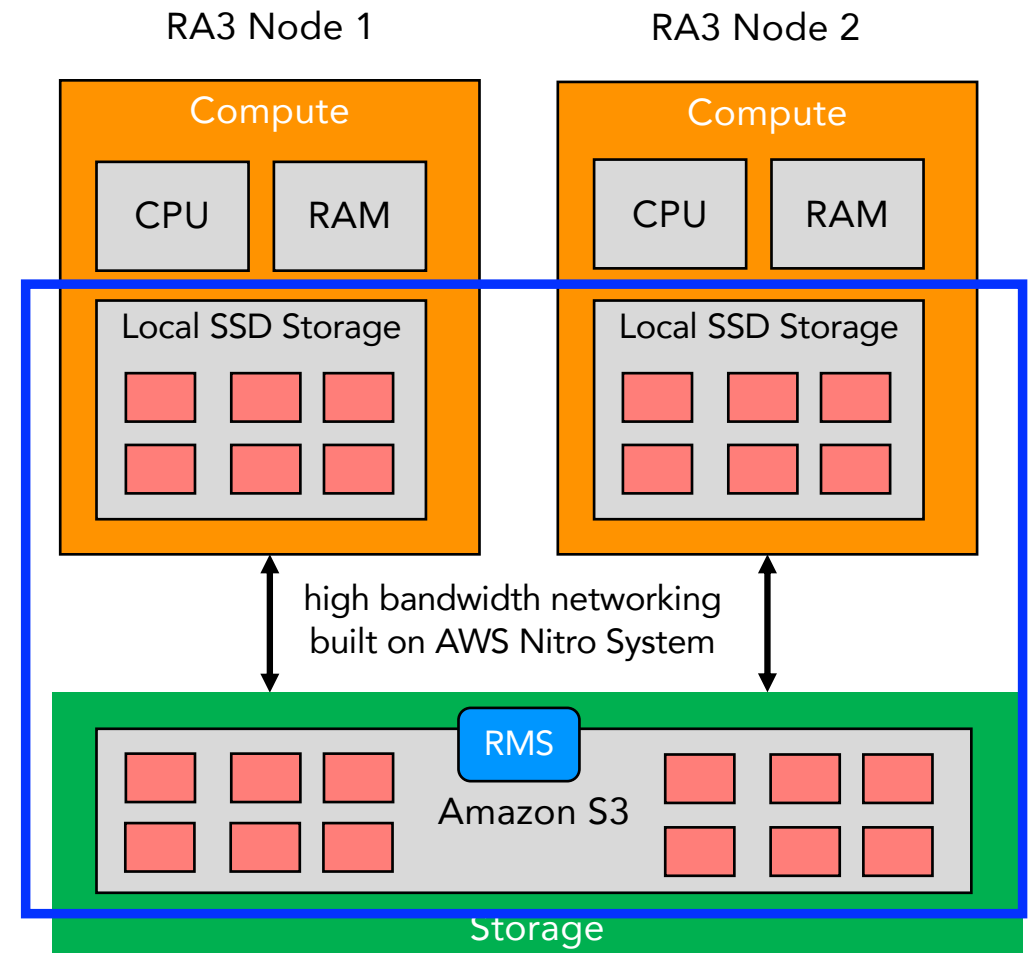
Nodes (in a Redshift Cluster)

- Redshift Nodes - i) Leader Node; ii) Compute Node
- What is a Node?



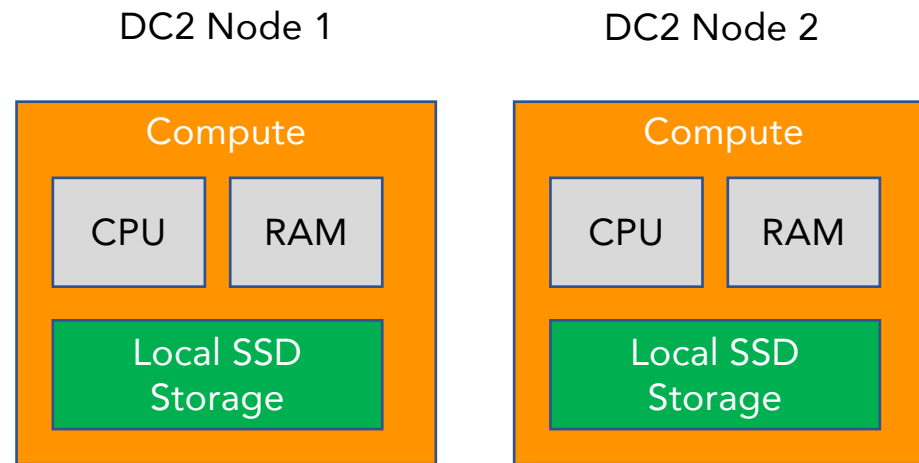
Node Types

- RA3
- SSD for hot data
- S3 for cold data
- Redshift Managed Storage
- Network
- Automatic data placement across tiered storage
 - Data block temperature
 - Data block age
 - Workload patterns



Node Types

- DC2
- DS2 (deprecated)

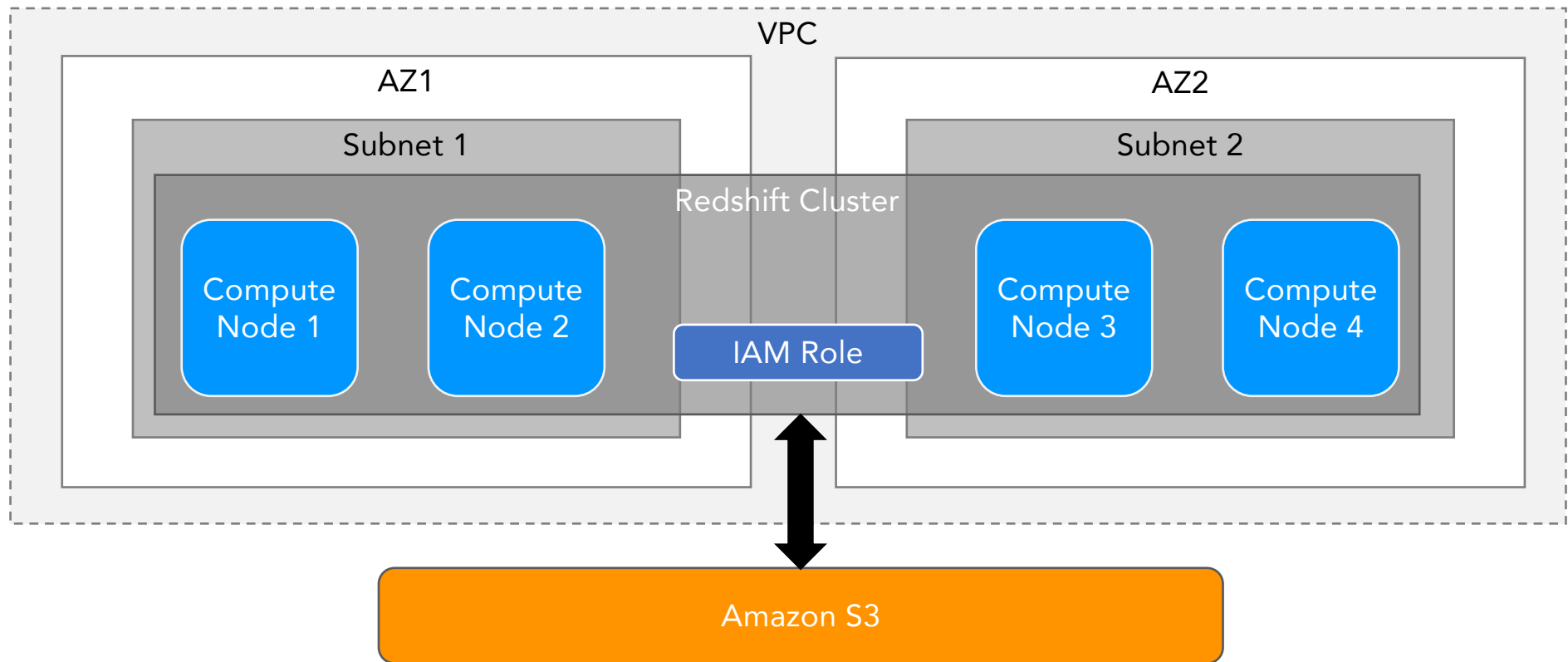


Node Types

RA3 node types						
Node type	vCPU	RAM (GiB)	Default slices per node	Managed storage limit per node ¹	Node range with create cluster	Total managed storage capacity ²
ra3.xlplus (single-node)	4	32	2	4 TB	1	4 TB ³
ra3.xlplus (multi-node)	4	32	2	32 TB	2–16 ⁴	1024 TB ⁴
ra3.4xlarge	12	96	4	128 TB	2–32 ⁵	8192 TB ⁵
ra3.16xlarge	48	384	16	128 TB	2–128	16,384 TB

Dense compute node types						
Node type	vCPU	RAM (GiB)	Default slices per node	Storage per node	Node range	Total capacity
dc2.large	2	15	2	160 GB NVMe-SSD	1–32	5.12 TB
dc2.8xlarge	32	244	16	2.56 TB NVMe-SSD	2–128	326 TB

Cluster Topology (Provisioned cluster)

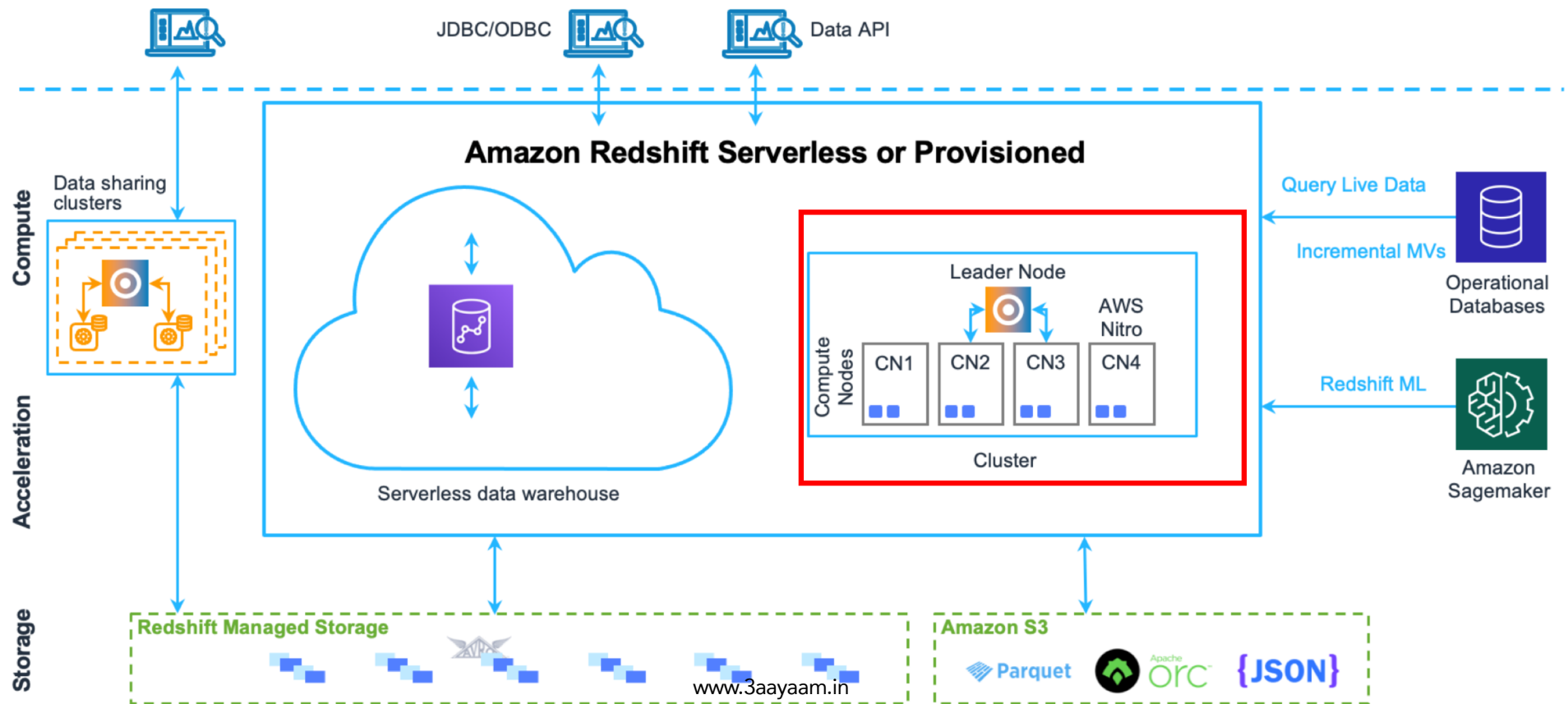


Create Cluster from Console

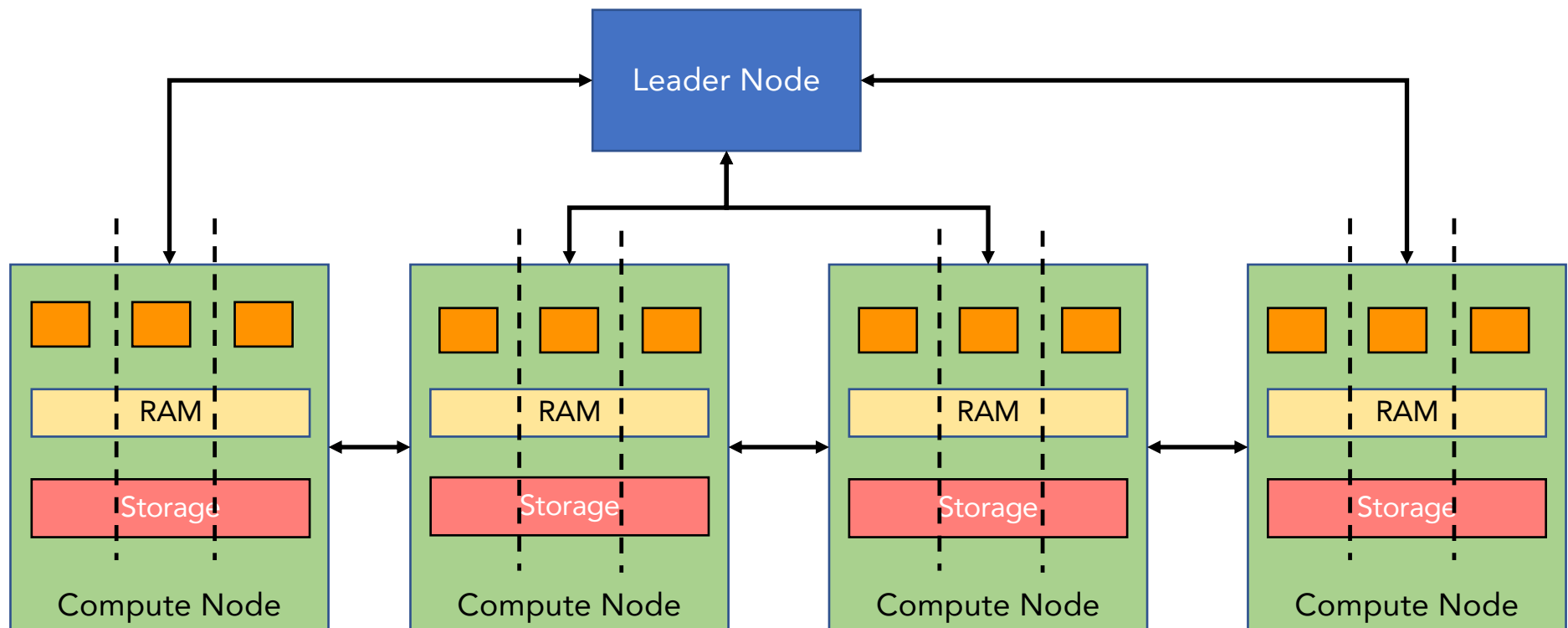
- Create VPC
- Create subnets
- Create security groups
- Create subnet group
- Create Redshift cluster

- Demo

Redshift Architecture



Redshift Architecture



Node Slices

- Portion of node's CPU and Memory
- Handles portion of disk space – subset of data

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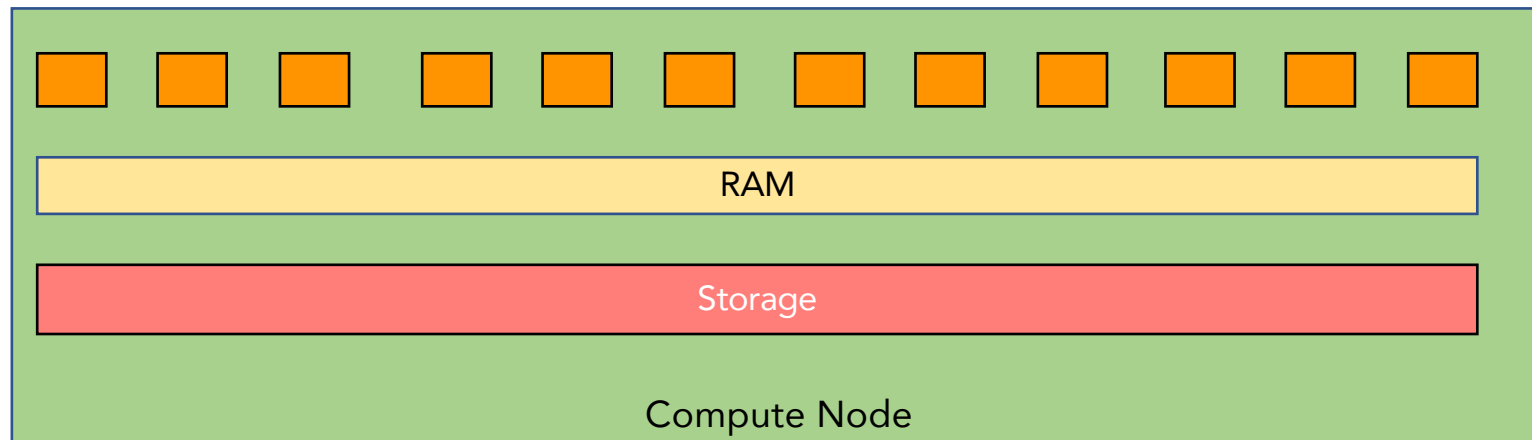
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MPP

- Fast run of complex queries operating on large amounts of data.
- Multiple compute nodes handle all query processing leading up to final result aggregation.
- Amazon Redshift distributes the rows of a table to the compute nodes so that the data can be processed in parallel.



Columnar Storage

Order_id	Customer_id	Seller_id	Order_date	Order_amount	Order_status	location
112ar56	223098	AA3-YID	2024-11-24	985.00	Pending	Bangalore
981cyr3	879223	RE4-C20	2024-12-08	11000.00	Delivered	Mumbai
625uidq	981090	NIT-004	2025-03-02	34600.00	Delivered	Mumbai

Row oriented storage

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Zone Maps

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345abc3	665431	XRA-98E	2025-05-01	9045.50	Pending	Delhi
843cxd2	874895	VTW-457	2024-06-20	9864.00	Pending	Bangalore
5478gt7	485694	HJD-327	2024-09-04	450.00	Processing	Bangalore
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Block 0 Min - 112ar56 Max - 981cyr3	Block 2 Min - 223098 Max - 981090	Block 4 Min - AA3-YID Max - XRA-98E	Block 6 Min - 2024-11-24 Max - 2025-03-02	Block 8 Min - 985.00 Max - 34600.00	Block 10 Min - Delivered Max - Pending	Block 12 Min - Bangalore Max - Mumbai
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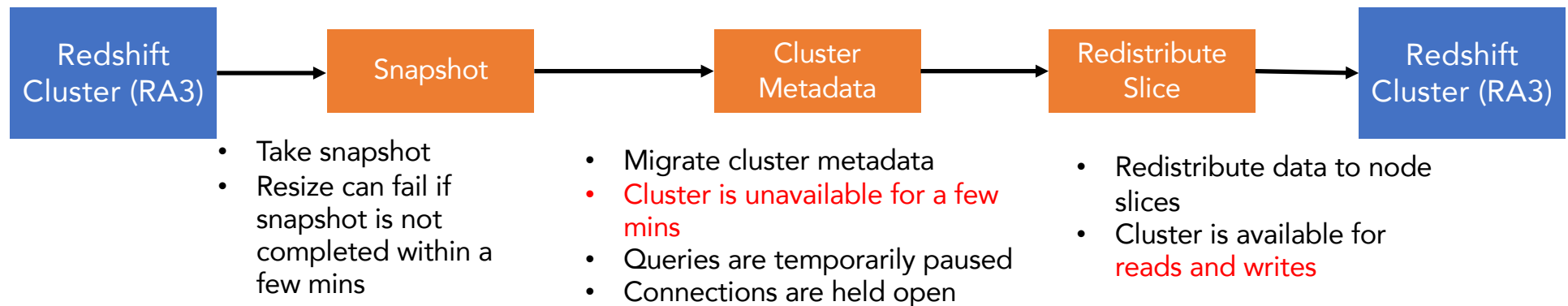
Redshift Cluster Management - Resize

- Elastic Resize

- Add nodes
- Add or remove nodes
- Change node type – From DC2 to RA3
- Completes quickly (within 10 mins)
- Data slices are redistributed

- Elastic Resize

- In-place resize when node type doesn't change, queries and sessions are held.
- New cluster is created when node type changes and queries are dropped



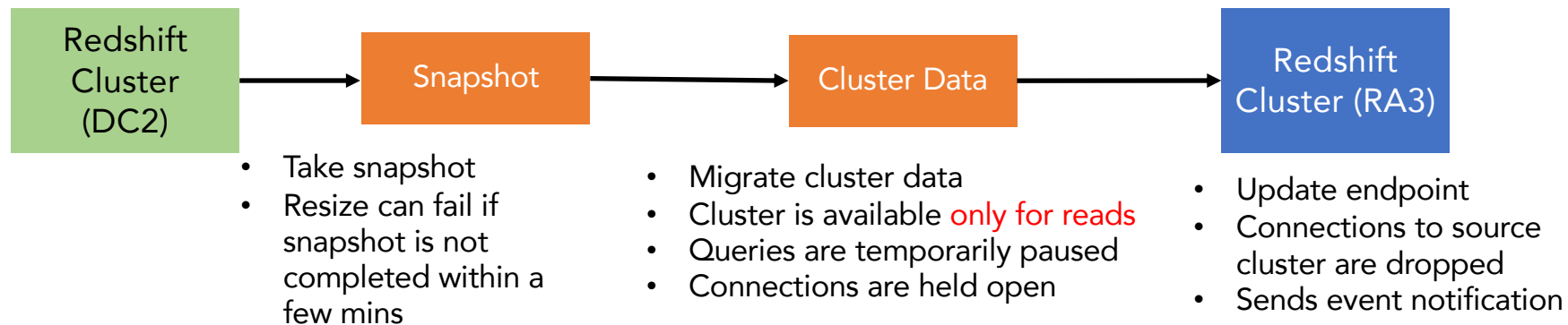
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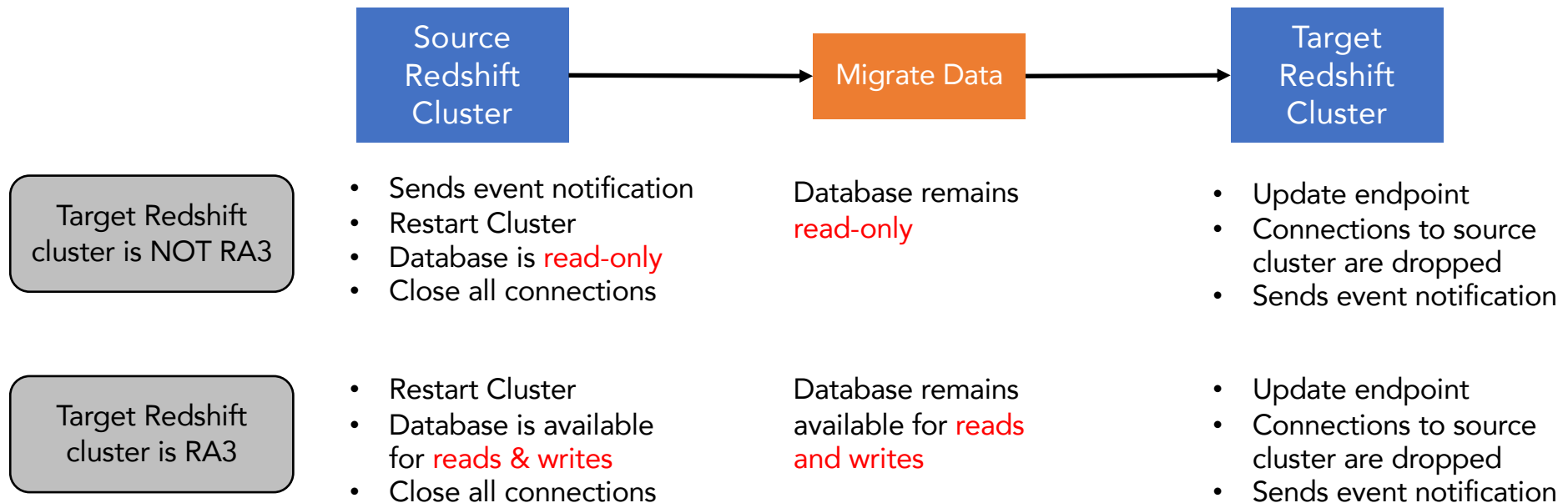
Redshift Cluster Management - Resize

Original node type	Growth limit	Reduction limit
ra3.16xlarge	4x (from 4 to 16 nodes, for example)	To one quarter of the number (from 16 to 4 nodes, for example)
ra3.4xlarge	4x	To one quarter of the number
ra3.xlplus	2x (from 4 to 8 nodes, for example)	To one quarter of the number
dc2.8xlarge	2x	To one half of the number (from 16 to 8 nodes, for example)
dc2.large	2x	To one half of the number

Redshift Cluster Management - Resize

- Classic Resize

- Resizing which are not supported by Elastic Resize
- Add or remove nodes



Demo

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Cluster Management – Pause & Rename

- Cluster can be paused and resumed.
- Compute billing is not charged.
- Storage cost is incurred.
- Steps
 - Creates snapshot
 - Terminates queries
- Rename cluster
 - Endpoint name changes
 - Cluster is not available until new name is effective
 - Cluster is rebooted
 - Connections are dropped

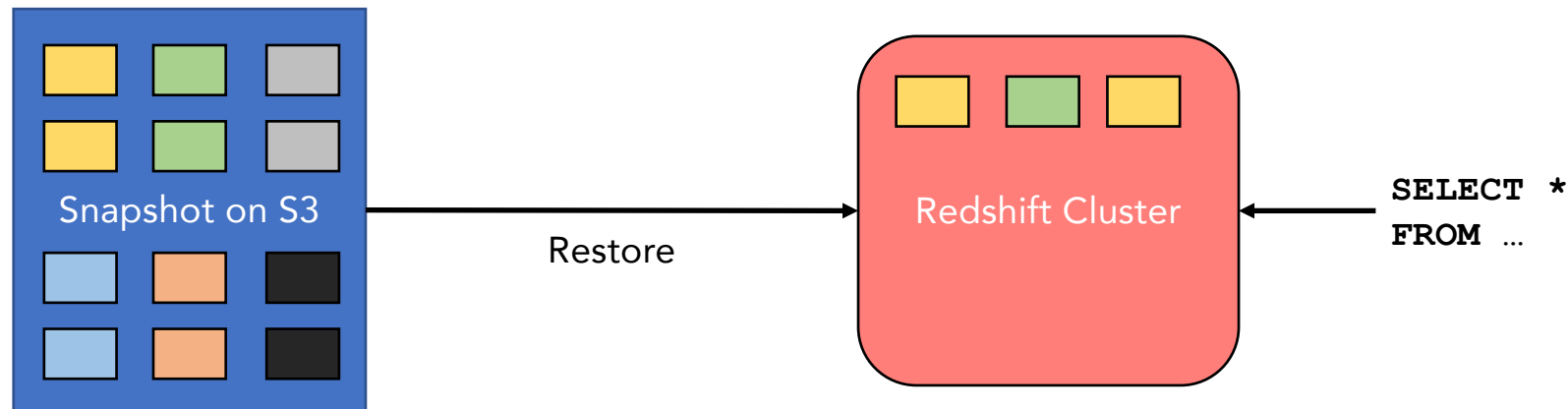
Demo

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Snapshot and Backup

- Automated snapshots
- Stored in S3
- Takes incremental snapshots that track changes in the cluster since the previous **automated snapshot**.
- Incremental snapshots at every 8 hrs or every 5 GB per node data changes
- Lazy loading after restore (to a new cluster)
- Manual snapshots
- Tables can be excluded from snapshots
- Automatic copy of snapshots (automatic and manual) to another region

Lazy Loading



- Cluster is available before all data is loaded.
- Streams data on demand from the snapshot in response to active queries
- Loads the remaining data in the background.

Demo

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Summary

- Cluster
- Nodes & node types (RA3, DC2)
- Redshift Managed Storage (in RA3 nodes)
- Created Redshift cluster
- Architecture
- Node slices
- Columnar storage
- Zone maps
- Cluster resize, pause, resume and drop
- Snapshot and backup